Q22 Work done is required on the loop to maintain uniform speed, when the right edge of the square loop is out of the magnetic field and only the left edge is in the field. This is because that there will be a resistive magnetic force acting on the left edge of the loop, and this occurs over a distance of L, the length of each side of the square loop.

Hence, total work done against resistive force,

$$
\begin{aligned}
W &= F_B L \\
&= (BI_{\text{induced}}L)(L) \\
&= B\left(\frac{E_{\text{induced}}}{R}\right)L^2 \\
&= B\left(\frac{BLv}{R}\right)L^2 \\
&= \frac{B^2L^3}{R}v
\end{aligned}
$$

where B is the flux density of the field, R is the resistance of each side of the wire.

Hence W is directly proportional to v.

Answer: B

Q23 Average e.m.f. induced